prompt:

Determine the least positive integer x for which the sum of x and 4609 gives a remainder of 2104 when divided by 12.

initial_reasoning_steps:

Let's think step by step.

Step 1: We know that $x + 4609 \equiv 2104 \pmod{12}$.

Step 2: We can rewrite the given congruence as $x \equiv 2104 - 4609 \pmod{12}$.

Step 3: Calculating $2104 - 4609 = -2505$.

Step 4:

chosen_step:

Now we need to find the equivalent positive integer for $-2505 \pmod{12}$.

To do this, we divide -2505 by 12 and find the remainder. $-2505 \div 12 = -208$ remainder -9.

However, we want a positive remainder, so we add 12 to -9 to get a positive remainder. $-9 + 12 = 3$.

So, $-2505 \equiv 3 \pmod{12}$.

rejected_step:

We want to find the positive integer x, so we add 12 until we get a positive number: $-2505 + 12 \cdot 210 = 30$.

Figure 1. An example of a Step-DPO preference data sample.